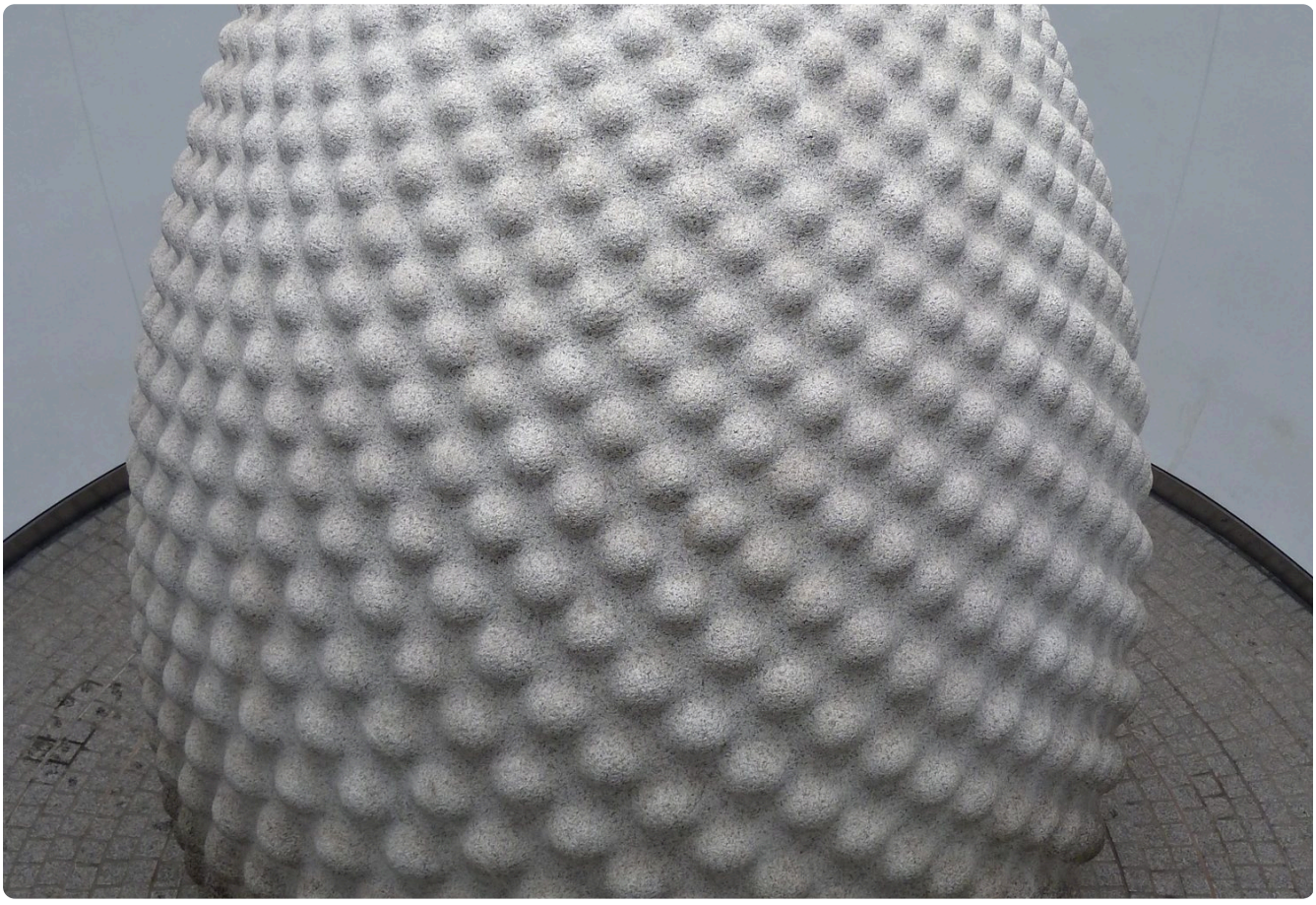
DESIGN COMPUTED · FREE GUIDE

The golden angle: the turn that fills every gap

The free guide to the reel on the Seed at the Eden Project: why plants turn each new seed by 137.5 degrees, what that does, and how to use the same pattern in your own designs.



How it works

A sunflower head packs as many seeds as it can into a round space. If each seed is turned by a simple fraction of a full turn (a quarter, a fifth), the seeds line up in spokes with gaps between them.

Turn each seed by about 137.5 degrees, the golden angle, and no spokes form. Every gap fills as the pattern grows, and you see spirals in two directions. Their counts are neighbouring Fibonacci numbers, such as 34 and 55 on a large sunflower head.

In 1979 Helmut Vogel described this with a simple model: seed number n sits at distance √n from the centre, turned n times 137.5 degrees from the first.

The Eden Project

At the Eden Project in Cornwall, the sculptor Peter Randall-Page carved Seed, a large granite sculpture whose surface follows the same spiral pattern. The building around it, the Core, has a roof developed by Grimshaw with the sculptor and the engineers; its design is described as inspired by phyllotaxis, the pattern of plant growth. The roof is inspired by the pattern; it is not a copy of Vogel's formula.

Built examples

Seed and the Core, Eden Project, Cornwall (2005)	Grimshaw with Peter Randall-Page: roof and sculpture based on phyllotaxis.
Vogel's model (1979)	A formula for seed positions that you can run in any spreadsheet.
Sunflower heads	Natural examples with 34 and 55 spirals.
Pine cones and pineapples	Spirals in two directions, with Fibonacci counts.

Try it yourself

1. Open a spreadsheet or the free tool Desmos.
2. Make a list of numbers n from 1 to 500.
3. Compute the distance r = √n and the angle a = n × 137.5°.

4 Convert to x = r cos(a) and y = r sin(a), then plot the points.

5 Change the angle to 137° and to 140°, and see how gaps and spokes appear.

6 Try 90° (a quarter turn) to see the spokes the golden angle avoids.

7 Use the pattern for a facade, a floor tile, a light fitting or a plan: each point can be a hole, a plant or a column.

The rule to remember

Next time you repeat a shape around a point, don't turn it by a round number.

Rounding the angle	Small changes matter: keep 137.5.
Copying the look only	Use the rule to place things, not just to decorate.
Forgetting size	Let elements grow with distance (√n) so they don't overlap.

Go further

Golden angle, Wikipedia
https://en.wikipedia.org/wiki/Golden_angle

The Algorithmic Beauty of Plants, chapter 4
https://algorithmicbotany.org/papers/abop/abop-ch4.pdf

How Eden's seed sculpture germinated, Building Design
https://www.bdonline.co.uk/how-edens-seed-sculpture-germinated/3118648.article

Seed at the Eden Project
https://edenproject.com/visit/things-to-do/seed

Credits
Photos: see the reel caption (Wikimedia Commons, free licences).
Facts: Wikipedia, algorithmicbotany.org, Eden Project, Building Design.

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